

1. Algorithm

1. Start.
2. Display the menu options to the user:
 - Choice 1: Count number of even digits and odd digits.
 - Choice 2: Find sum of even digits and sum of odd digits separately.
 - Choice 3: Exit.
3. Read the user's choice.
4. If Choice is 1 or 2:
 - Prompt and input a positive integer num.
5. Execute operations based on choice using switch-case:
 - **Case 1 (Count Digits):**
 - Initialize `even_dig = 0`, `odd_dig = 0`, and `temp = num`.
 - Loop while `temp > 0`:
 - Extract last digit: `rem = temp % 10`.
 - If `rem % 2 == 0`, increment `even_dig`.
 - Else, increment `odd_dig`.
 - Update `temp = temp / 10`.
 - Display `even_dig` and `odd_dig`.
 - **Case 2 (Sum of Digits):**
 - Initialize `sum_even = 0`, `sum_odd = 0`, and `temp = num`.
 - Loop while `temp > 0`:
 - Extract last digit: `rem = temp % 10`.
 - If `rem % 2 == 0`, `sum_even += rem`.
 - Else, `sum_odd += rem`.
 - Update `temp = temp / 10`.
 - Display `sum_even` and `sum_odd`.
 - **Case 3:** Exit program.
 - **Default:** Display "Invalid Choice!".
6. Stop.

2. Test Case Table

Test Case ID	Choice	Input Number	Expected Output	Actual Output	Status
TC01	1	12045	Even_dig = 3, Odd_dig = 2	Even_dig = 3, Odd_dig = 2	Pass
TC02	2	12345	Sum_even = 6, Sum_odd = 9	Sum_even = 6, Sum_odd = 9	Pass
TC03	1	2468	Even_dig = 4, Odd_dig = 0	Even_dig = 4, Odd_dig = 0	Pass
TC04	2	1357	Sum_even = 0, Sum_odd = 16	Sum_even = 0, Sum_odd = 16	Pass
TC05	3	N/A	Exiting Program.	Exiting Program.	Pass

3. C Program Source Code

```
#include <stdio.h>

int main() {
    int choice;
    long long num, temp;
    int rem;

    printf("--- MENU ---\n");
    printf("1. Count number of even digits and odd digits\n");
    printf("2. Find sum of even digits and sum of odd digits\n");
    printf("3. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice) {
        case 1: {
            int even_dig = 0, odd_dig = 0;
            printf("Enter a positive integer: ");
            scanf("%lld", &num);

            temp = num;
            if (temp == 0) {
                even_dig = 1;
            } else {
                while(temp > 0) {
                    rem = temp % 10;
                    if(rem % 2 == 0)
                        even_dig++;
                }
            }
        }
    }
}
```

```

        else
            odd_dig++;
            temp = temp / 10;
        }
    }
    printf("Number: %lld", num);
    printf("Even_dig = %d, Odd_dig = %d", even_dig, odd_dig);
    break;
}
case 2: {
    int sum_even = 0, sum_odd = 0;
    printf("Enter a positive integer: ");
    scanf("%lld", &num);

    temp = num;
    while(temp > 0) {
        rem = temp % 10;
        if(rem % 2 == 0)
            sum_even += rem;
        else
            sum_odd += rem;
        temp = temp / 10;
    }
    printf("Number: %lld", num);
    printf("Sum_even = %d, Sum_odd = %d", sum_even, sum_odd);
    break;
}
case 3:
    printf("Exiting program.");
    break;
default:
    printf("Invalid choice!");
}

return 0;
}

```

4. Execution & Output (Kali Linux Terminal)

```

user@kali:~$ gcc program.c -o program
user@kali:~$ ./program
--- MENU ---
1. Count number of even digits and odd digits
2. Find sum of even digits and sum of odd digits
3. Exit
Enter your choice: 1

```

Enter a positive integer: 12045

Number: 12045

Even_dig = 3, Odd_dig = 2

user@kali:~\$./program

--- MENU ---

1. Count number of even digits and odd digits
2. Find sum of even digits and sum of odd digits
3. Exit

Enter your choice: 2

Enter a positive integer: 12345

Number: 12345

Sum_even = 6, Sum_odd = 9